

Graph Introduction & Terminologies Data Structures

> **Definition:** A **Graph** is a non-linear data structure denoted as $G = (V, E)$, consisting of a non-empty finite set of **Vertices** (Nodes) V and a set of **Edges** (Arcs) E connecting pairs of vertices.

1. Detailed Technical Explanation

Core Graph Terminologies:

1. **Vertex (Node):** An individual data point or entity in the graph.
2. **Edge:** A connection between two vertices.

2. Memory Keywords & Properties

- **Vertices & Edges:** $G = (V, E)$.
- **Hamiltonian Cycle:** A cycle that visits every vertex exactly once.

3. Must-Write Points for Exams

- In an undirected complete graph with N vertices, the maximum number of edges is $N(N - 1) / 2$.

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4. Quick Recall Flow

$G = (V$